BANKING CUSTOMER ANALYTICS

Exploratory Data Analysis & Business Insights

Data Analytics Portfolio Project | 3,000 Client Records | Power BI + Python + Excel

3,000 Total Clients Analysed	**$171K** Avg. Estimated Income	**$671K** Avg. Bank Deposits	**$591K** Avg. Bank Loans
4 Loyalty Tiers	**5** Nationalities	**3** Fee Structure Tiers	**100%** Dataset Completeness

1. Business Problem Statement

Modern retail banks manage thousands of diverse client relationships — each with distinct financial behaviours, risk profiles, and loyalty patterns. Without a structured analytical framework, institutions risk misallocating resources, designing one-size-fits-all products, and failing to retain high-value customers.

1.1 Problem Context

This project investigates the following core business challenge:

> *"How can a bank leverage customer demographic, behavioural, and financial data to better understand client segments, identify high-value customers, manage risk exposure, and design targeted product offerings?"*

1.2 Business Questions Addressed

- **Customer Profiling:** Who are the bank's most profitable and loyal customers?
- **Loyalty Drivers:** What demographic and financial factors drive loyalty tier classification?
- **Income-Behaviour Linkage:** How does income level relate to savings, borrowing, and investment behaviour?
- **Risk Segmentation:** Which customer segments carry the highest financial risk to the bank?
- **Product Distribution:** Are there patterns in product adoption (credit cards, loans, deposits) across nationalities?
- **Fee Optimisation:** How can the bank better target fee structure offerings to the right customers?

1.3 Business Objectives

The analysis is designed to support three strategic objectives:

- **Segmentation:** Enable data-driven customer segmentation to improve targeted marketing and personalised service delivery.
- **Retention:** Identify at-risk and high-value customer groups to improve retention strategies and reduce churn.
- **Risk & Pricing:** Provide a foundation for risk-based pricing and fee structure optimisation across client tiers.

2. Dataset Overview

The Banking Customer Dataset comprises 3,000 anonymised client records collected across multiple geographic regions. The dataset is clean — containing no missing values — and spans 25 feature columns covering demographic, financial, and product-related attributes.

2.1 Dataset Snapshot

Attribute	Details	Category
Total Records	3,000 clients	Scope
Total Features	25 columns	Coverage
Missing Values	None (100% complete)	Data Quality
Age Range	17 to 85 years	Demographics
Nationality Groups	5 (European, Asian, American, Australian, African)	Demographics
Loyalty Tiers	Jade, Silver, Gold, Platinum	Classification
Fee Structures	High, Mid, Low	Products

2.2 Feature Categories

The 25 columns span four distinct domains:

- **Demographic:** Client ID, Name, Age, Gender (GenderId), Nationality, Occupation, Location ID
- **Relationship:** Joined Bank date, Banking Contact, Loyalty Classification, Fee Structure
- **Financial:** Estimated Income, Superannuation Savings, Credit Card Balance, Bank Loans, Bank Deposits, Checking Accounts, Saving Accounts, Foreign Currency Account, Business Lending, Properties Owned, Amount of Credit Cards
- **Risk & Classification:** Risk Weighting, BRId, IAId

3. Methodology & Tools

3.1 Technology Stack

Tool	Purpose	Application in Project
Python (Pandas)	Data Wrangling	Loading, cleaning, transforming, and feature engineering (Income Band)
Seaborn / Matplotlib	Visualisation	Histograms, countplots, heatmaps, distribution plots
NumPy	Computation	Numerical operations and array processing
Power BI	Business Dashboard	Interactive visuals: KPI cards, bar charts, slicers, trend analysis
Jupyter Notebook	EDA Environment	Step-by-step exploratory workflow with inline charts

3.2 Analytical Approach

The analysis followed a structured Exploratory Data Analysis (EDA) pipeline:

- **Step 1 — Data Understanding:** Data loading, shape inspection (3000 rows x 25 cols), dtypes review, and null-value check.
- **Step 2 — Feature Engineering:** Income banding: Estimated Income segmented into Low (<$100K), Medium ($100K–$300K), and High (>$300K) bands.
- **Step 3 — Univariate Analysis:** Frequency distributions, value counts, and bar charts for all categorical variables including loyalty tier, fee structure, nationality, and risk weighting.
- **Step 4 — Bivariate Analysis:** Countplots segmented by nationality for each categorical variable to detect inter-group behavioural differences.
- **Step 5 — Numerical Distributions:** Histograms with KDE curves for Age, Income, Superannuation, Credit Card Balance, Loans, Deposits, and Savings.
- **Step 6 — Correlation Analysis:** Pearson correlation heatmap across 8 financial variables to identify linear dependencies between monetary features.
- **Step 7 — Dashboard & Reporting:** Power BI dashboard built for executive-level summary with slicers for loyalty tier, nationality, and fee structure.

4. Key Findings & Insights

4.1 Customer Demographics

Metric	Value / Insight
Average Customer Age	51 years (range: 17–85)
Largest Nationality Group	European — 1,309 clients (43.6%)

Second Largest Group	Asian — 754 clients (25.1%)
Smallest Nationality Group	African — 176 clients (5.9%)
Most Common Income Band	Medium ($100K–$300K) — 1,517 clients (50.6%)
High Income Band (>$300K)	456 clients (15.2%) — prime upsell targets

4.2 Loyalty Classification Distribution

The bank's customer base is predominantly classified at the entry Jade tier, with relatively few achieving Platinum status. This creates a significant upselling opportunity across Silver and Gold tiers.

Jade	Silver	Gold	Platinum
1,331 clients	**767 clients**	**585 clients**	**317 clients**
44.4%	25.6%	19.5%	10.6%

4.3 Financial Behaviour Insights

Analysis of financial variables revealed several meaningful patterns across the client base:

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Deposit-Savings Correlation (r = 0.75)
The strongest relationship in the dataset: clients with higher saving account balances consistently maintain larger total bank deposits. This suggests a conservative, security-driven financial profile for the majority of clients.

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Business Lending Links to Deposits & Loans
Business Lending shows moderate positive correlations with Bank Loans (r = 0.42) and Bank Deposits (r = 0.44). Businesses that actively borrow also tend to maintain larger deposit reserves, indicating commercially active client clusters.

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Income Drives Multi-Product Uptake
Estimated Income correlates moderately with Superannuation Savings (r = 0.37), Credit Card Balance (r = 0.30), and Business Lending (r = 0.33). Higher-income clients are the bank's most multi-product engaged segment.

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Credit Card Penetration
64.1% of clients hold a single credit card, 25.5% hold two, and 10.4% hold three. Multi-card customers are strong candidates for premium or co-branded card offerings.

4.4 Risk Profile Distribution

Risk Weighting ranges from 1 (lowest) to 5 (highest). The majority of clients fall in the low-to-moderate risk band, while a meaningful minority carry elevated risk:

Risk 1 (Low)	Risk 2	Risk 3	Risk 4	Risk 5 (High)

836 (27.9%)	1,222 (40.7%)	460 (15.3%)	322 (10.7%)	160 (5.3%)

Over 68% of clients hold Risk Weighting 1 or 2 — representing a stable, low-risk majority. However, the 482 clients with Risk 4 or 5 (16% of base) warrant active monitoring for credit exposure management.

4.5 Fee Structure & Loyalty Cross-Analysis

Cross-tabulation of Loyalty Classification against Fee Structure reveals that High fee clients dominate across all loyalty tiers — including 668 Jade-tier clients on High fees. This suggests a potential misalignment between fee paid and value received, pointing to fee optimisation opportunities.

Loyalty Tier	High Fee	Mid Fee	Low Fee
Jade	668	423	240
Silver	364	250	153
Gold	304	173	108
Platinum	140	116	61

5. Business Recommendations

Based on the EDA findings, the following strategic recommendations are proposed for the bank's product, marketing, and risk teams:

#	Area	Recommendation
01	**Loyalty Upselling**	Launch targeted campaigns for the 767 Silver-tier clients to progress to Gold. Focus incentives on deposit growth and product diversification, which correlate with higher loyalty tiers.
02	**Fee Realignment**	Review the 668 High-fee Jade clients — these clients are paying premium rates but receiving entry-tier service. Offering tailored mid-tier migration pathways could improve satisfaction and reduce churn risk.
03	**High-Income Targeting**	The 456 High-income clients ($300K+) are under-represented in the Platinum tier. Dedicated relationship managers and premium product bundles (wealth management, business lending, FX accounts) should be offered proactively.
04	**Risk Monitoring**	The 482 clients with Risk Weighting 4 or 5 should be flagged for quarterly credit review. Automated alerts linked to credit card balance thresholds and loan-to-income ratios would improve early warning capability.
05	**Multi-Product Cross-sell**	Clients with 2+ credit cards and existing business lending represent an ideal audience for structured investment products. Cross-selling superannuation top-ups and FX accounts to this cohort could increase revenue per client.

06	**Nationality Segmentation**	With Europeans forming 43.6% of the base, bespoke regional communication strategies — particularly for the smaller African (5.9%) and Australian (8.5%) segments — may improve engagement and product adoption.

6. Conclusions

This Exploratory Data Analysis of the Banking Customer Dataset demonstrates how structured data analysis can surface actionable insights from raw client records. Key takeaways from the project:

- The dataset is highly clean and well-structured, making it an ideal candidate for further predictive modelling (e.g., churn prediction, credit scoring).
- There is a clear and measurable relationship between income, savings behaviour, and product engagement — confirming the value of income-based customer segmentation.
- The Loyalty Classification system broadly aligns with financial capacity, but fee structures are not always proportional — representing a near-term optimisation opportunity.
- Risk is concentrated among a manageable minority, but early-warning monitoring systems would significantly improve credit portfolio health.
- Power BI dashboards provide the business with an accessible, real-time view of the client base — enabling data-driven decision-making without requiring technical expertise.

Future analytical work could include supervised machine learning to predict loyalty tier upgrades, customer lifetime value modelling, and time-series analysis of deposit and loan trends post client onboarding.

Appendix — Project Artefacts

A. Files Submitted

File	Format	Description
Banking_Dataset.csv	.csv	Raw customer dataset — 3,000 rows, 25 columns
Python_EDA.ipynb	.ipynb (Jupyter)	Full EDA notebook with code, charts, and inline insights
project_dashboard.pbix	.pbix (Power BI)	Interactive business dashboard with KPIs and slicers
Banking_Analytics_Report.docx	.docx	This report — project overview, findings, and recommendations

B. EDA Techniques Applied

- Descriptive statistics (mean, std, min/max, quartiles) via df.describe()
- Null-value audit via df.isnull().sum()
- Feature engineering: Income Band (Low / Medium / High) via pd.cut()
- Univariate visualisation: countplots for all categorical variables
- Bivariate visualisation: countplots with hue='Nationality' for cross-group comparison
- Numerical distributions: seaborn histplots with KDE overlay for 7 financial variables
- Correlation heatmap: Pearson correlation matrix for 8 financial features
- Cross-tabulation: Loyalty Classification vs Fee Structure via pd.crosstab()

— End of Report —